GenepoweRx® MultiForm Website

Genetic Science powered by K&H,

TECHNICAL DOCUMENTATION & WORKING PROOF

Project	GenepoweRx MultiForm Website
Version	v3.0 — Final Release
Date	March 2026
Tech Stack	React.js + Node.js + Express + MongoDB
Database	genepowerx_internals
Patient ID Format	KHGENEPOWERX-[PATIENTNAME]
Status	LIVE — Running on localhost:3000

This document contains the complete technical documentation of the GenepoweRx MultiForm Website including system architecture, features, API reference, database schema, and working proof screenshots of the live application.

1. PROJECT OVERVIEW

What is GenepoweRx MultiForm Website?

GenepoweRx MultiForm Website is a full-stack clinical genomics patient management system that digitises the entire patient intake workflow for the GenepoweRx genetic testing laboratory. It eliminates all paper forms by capturing patient data electronically through 5 specialised forms, generates professional branded PDFs automatically, and maintains a complete multi-visit history for every patient.

Key Highlights

5 Digital Forms	Lifestyle, Short Lifestyle, WES, ONCO, and Consent — each with full validation, draft save, and file upload
Auto Patient ID	KHGENEPOWERX-PATIENTNAME — unique ID auto-assigned on first submission
Multi-Visit System	Every re-submission of any form creates a new visit (Visit 1, 2, 3...) without overwriting previous data
PDF Generation	Professional branded PDFs auto-generated on every submission with GenepoweRx logo and brand colours
Digital Signature	Consent form includes HTML5 canvas signature pad — signature embedded in PDF
Patient Revisit	Returning patients enter their Patient ID to retrieve all records and start a new visit with pre-filled data
Excel Export	4 export types: All Patients, Form-wise, Clinical Insights, Single Patient
Draft System	Save partially filled forms and resume later using Draft ID

Technology Stack

Frontend	React 18, React Router 6, Recharts, React Hot Toast, React Icons
Backend	Node.js + Express 4 + Mongoose 8
Database	MongoDB — genepowerx_internals
PDF Engine	PDFKit (server-side) — logo + brand colours embedded
Excel	ExcelJS — styled multi-sheet workbooks
File Upload	Multer — Windows-compatible (copyFileSync + unlinkSync)
Port	Backend: 5000 | Frontend: 3000

2. APPLICATION SCREENSHOTS — WORKING PROOF

The following screenshots demonstrate the GenepoweRx MultiForm Website running live on localhost:3000. All features shown are fully functional with real patient data in the MongoDB database.

SCREEN 1 Clinical Dashboard

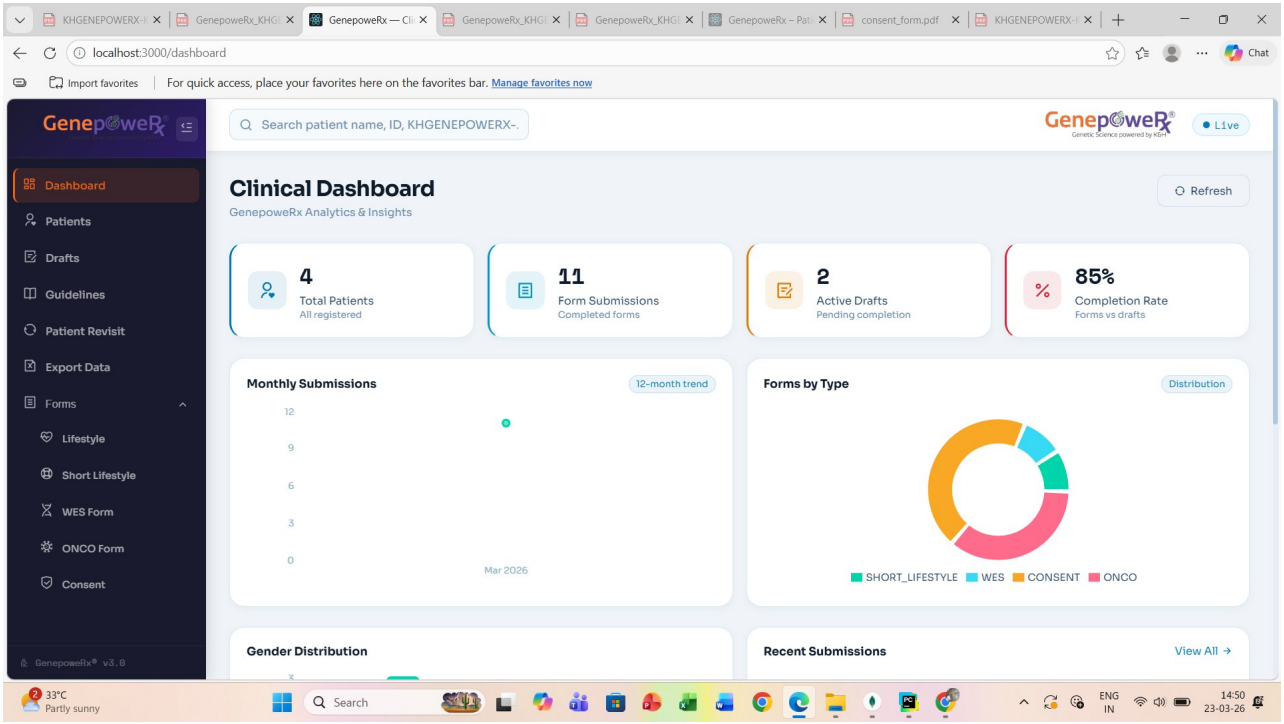

Fig. 1: The main dashboard shows real-time analytics: 4 registered patients, 11 form submissions, 2 active drafts, and 85% completion rate. Monthly submission trends and form distribution charts are displayed.

SCREEN 2 Patients List

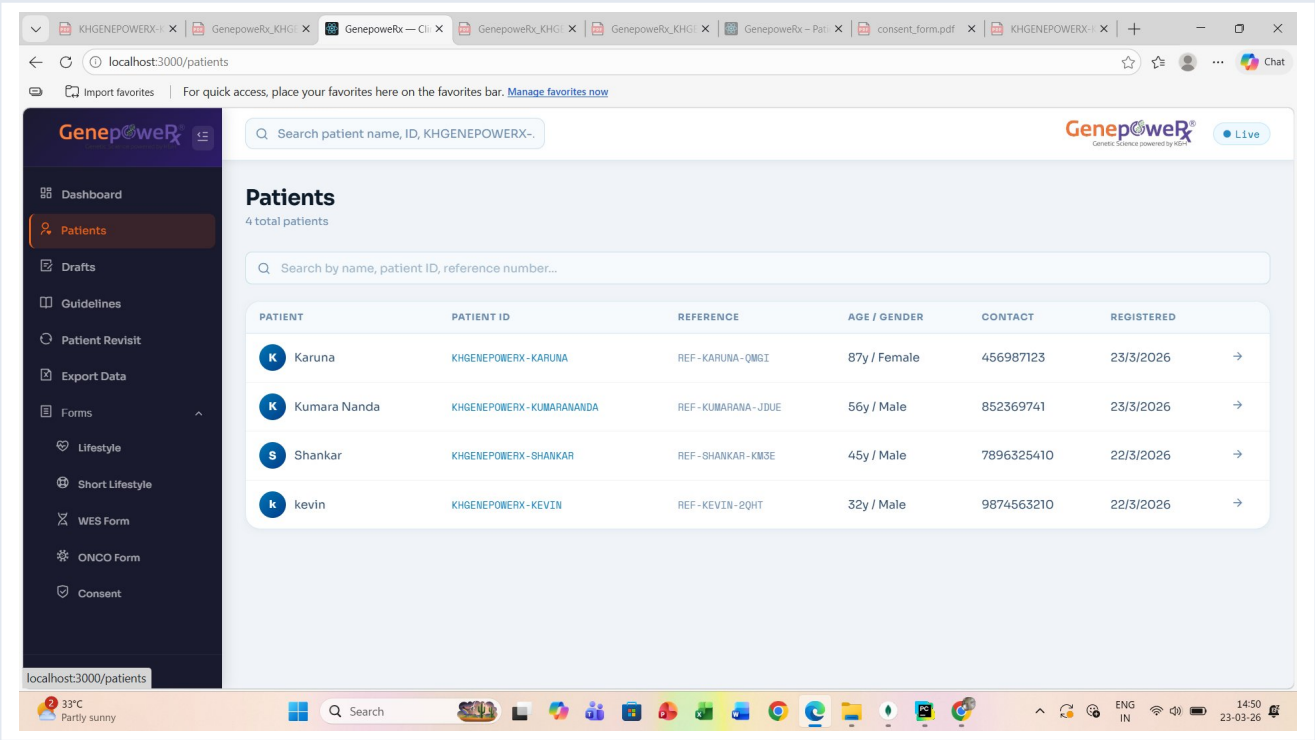

Fig. 2: *The Patients page lists all registered patients with their auto-generated Patient IDs (KHGENEPOWERX-NAME), Reference Numbers, age, gender, and registration date. Currently shows 4 patients: Karuna, Kumara Nanda, Shankar, and Kevin.*

SCREEN 3 Drafts Management

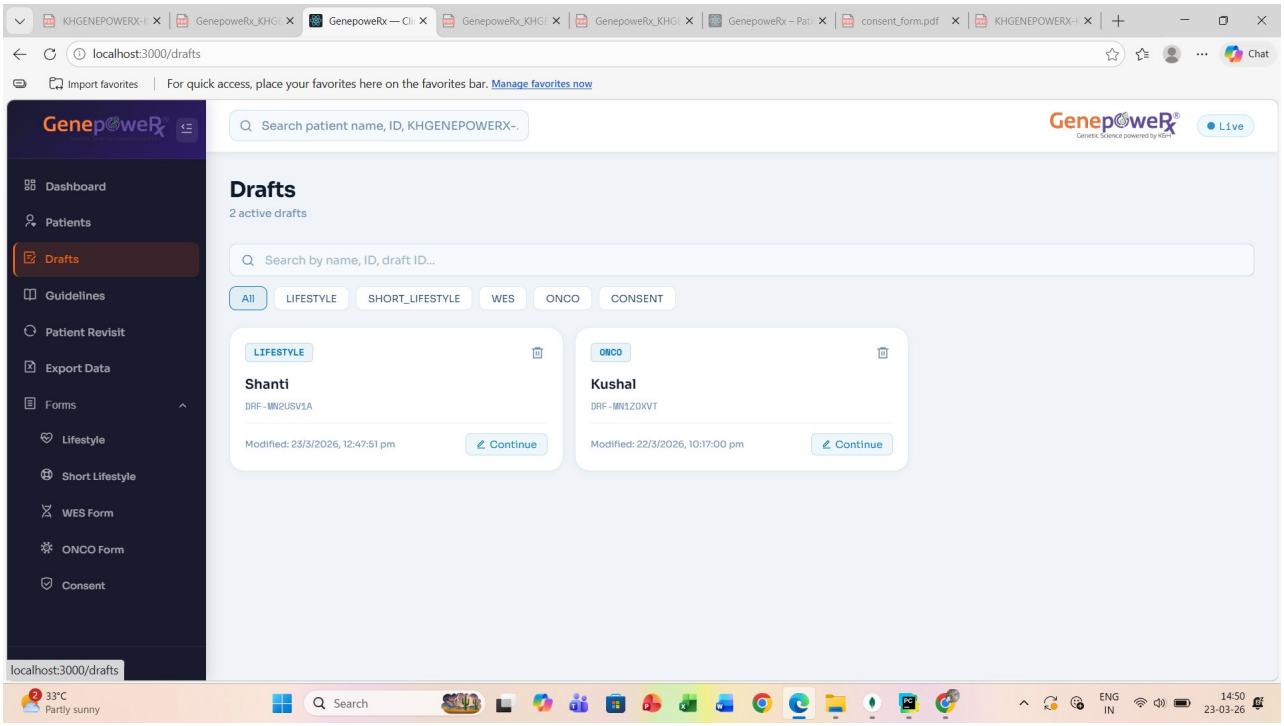

Fig. 3: The Drafts page shows 2 active drafts — one Lifestyle form for "Shanti" (DRF-MN2USV1A) and one ONCO form for "Kushal" (DRF-MN1Z0XVT). Drafts can be filtered by form type and resumed with the Continue button.

SCREEN 4 Patient ID Lookup — Search

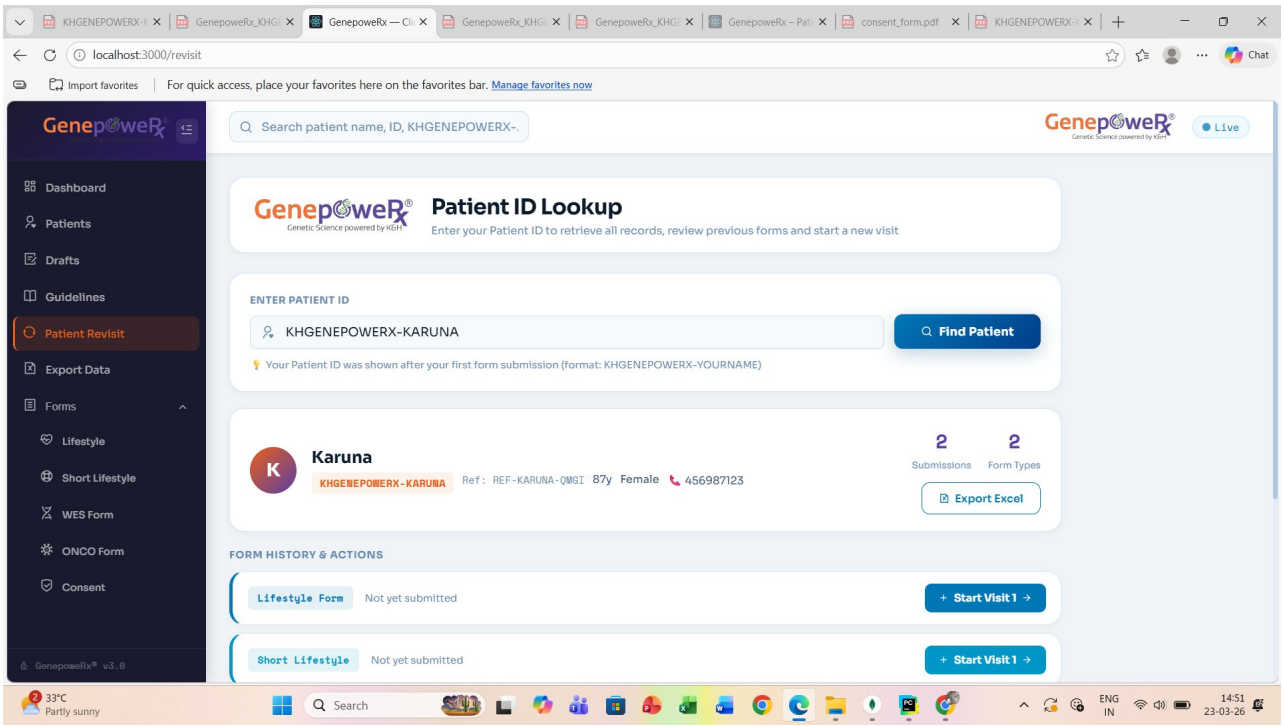

Fig. 4: The Patient Revisit page allows returning patients to enter their Patient ID and retrieve all previous records. Here, patient "Karuna" (KHGENEPOWERX-KARUNA) has been found with 2 submissions across 2 form types.

SCREEN 5 Patient ID Lookup — Form History

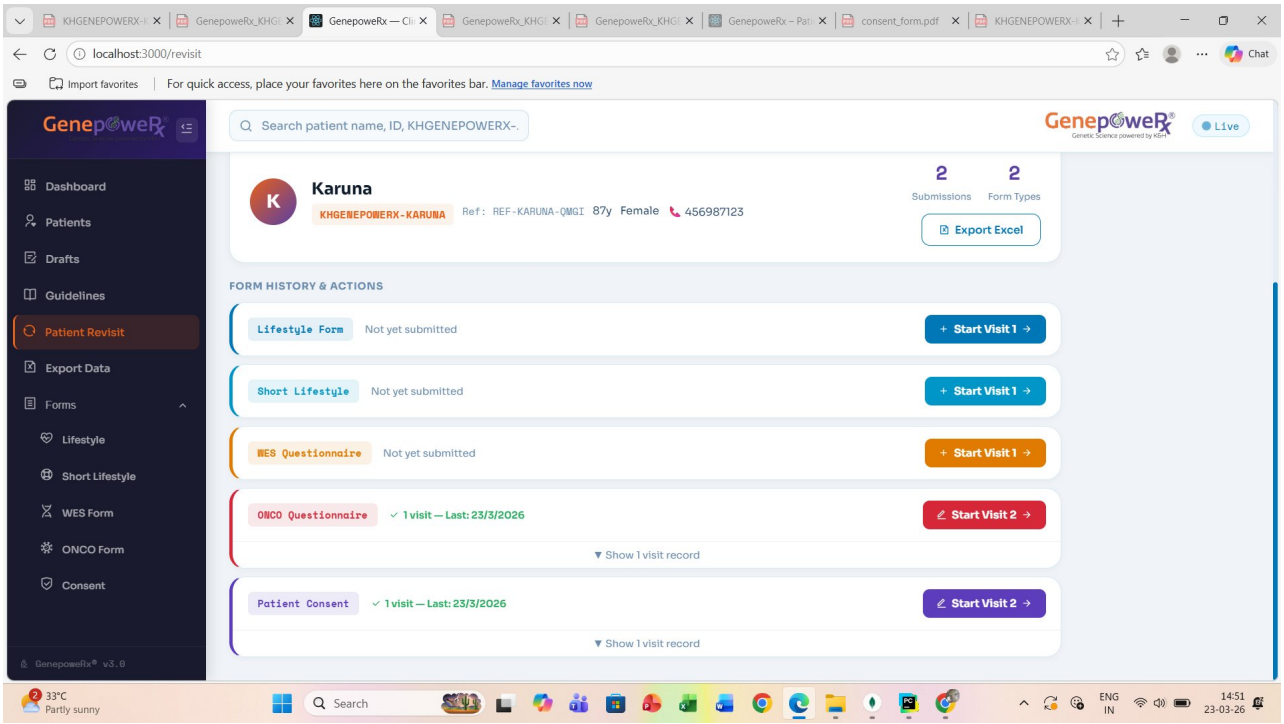

Fig. 5: After lookup, the full form history is shown for patient Karuna. ONCO Questionnaire and Patient Consent both have 1 visit (last: 23/3/2026) with "Start Visit 2" buttons. Lifestyle, Short Lifestyle, and WES forms show "Start Visit 1" as not yet submitted.

SCREEN 6 Data Export — All Patients & Form-wise

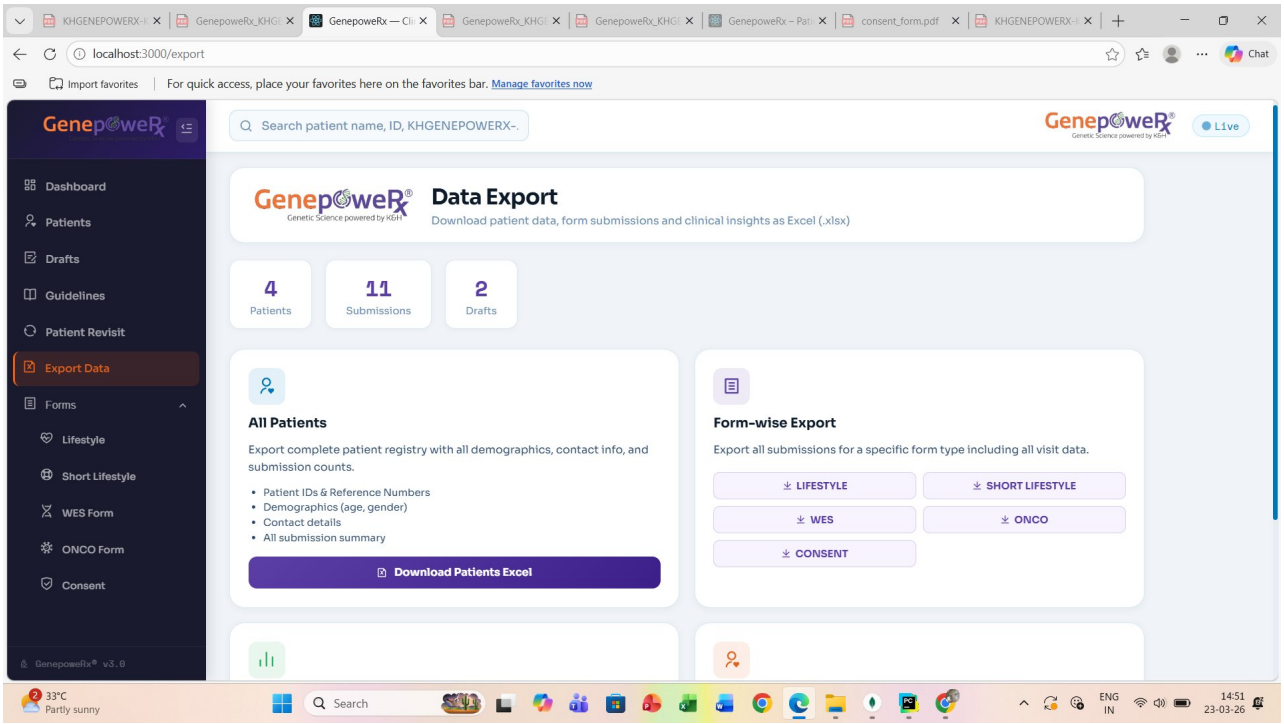

Fig. 6: *The Export Data page provides 4 export types. "All Patients" downloads a complete patient registry Excel file. "Form-wise Export" allows downloading submissions per form type: LIFESTYLE, SHORT LIFESTYLE, WES, ONCO, or CONSENT.*

SCREEN 7 Data Export — Clinical Insights & Single Patient

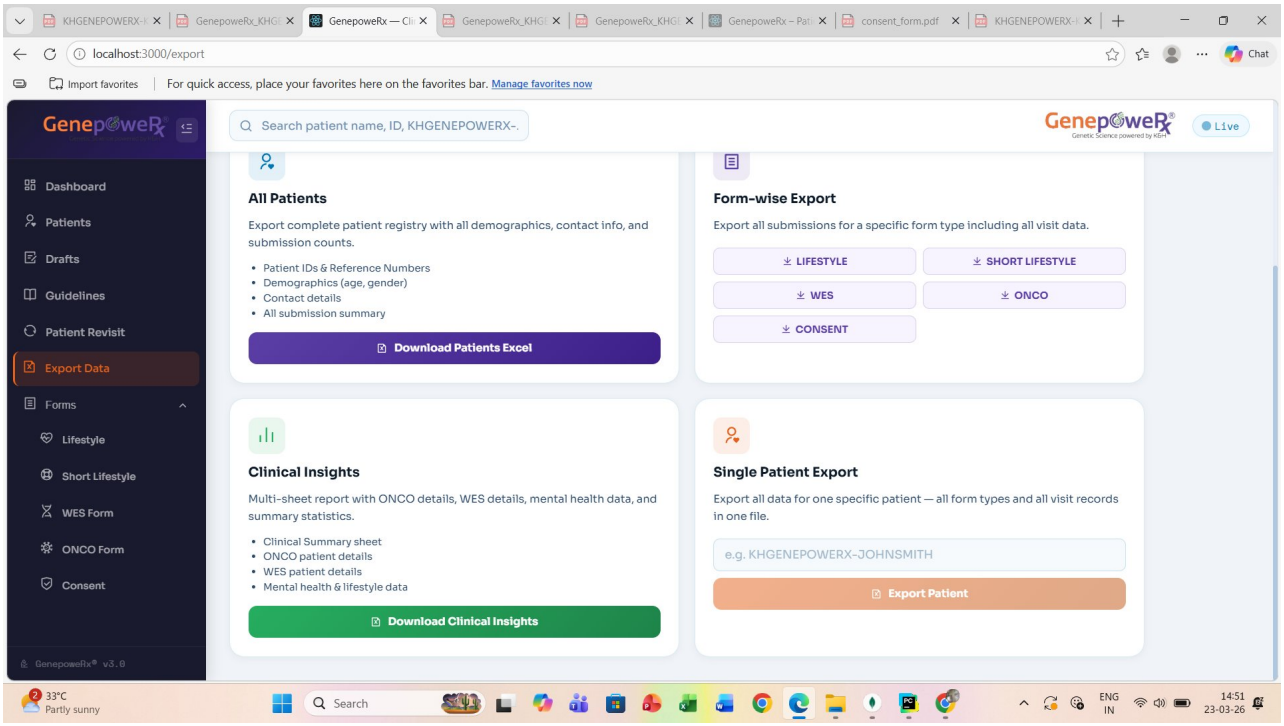

Fig. 7: The lower half of Export Data shows "Clinical Insights" (multi-sheet Excel with ONCO details, WES details, mental health summary) and "Single Patient Export" where a specific Patient ID can be entered to download all their data.

SCREEN 8 Lifestyle Patient Intake Form

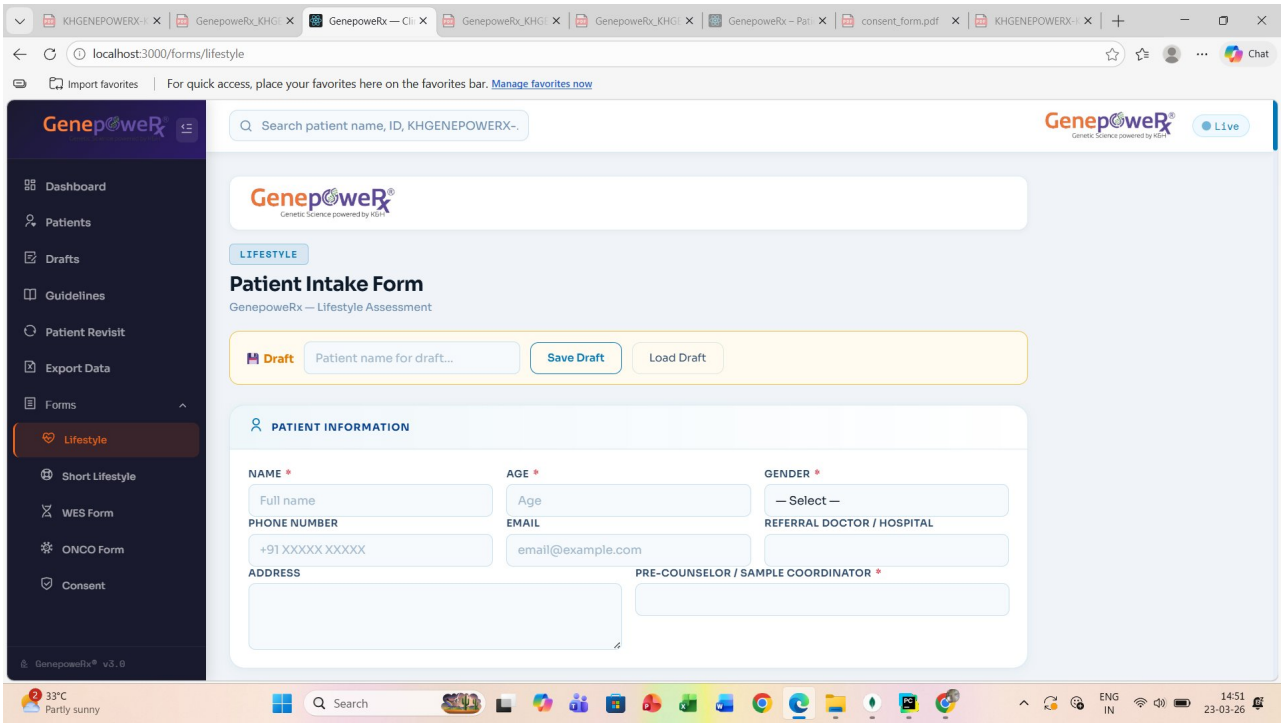

Fig. 8: *The Lifestyle Form is the primary patient intake form with GenepoweRx branding. It includes a Draft Bar for saving progress, and sections for Patient Information (Name, Age, Gender, Phone, Email, Address, Referral Doctor, Coordinator).*

SCREEN 9 WES Condition Specific Questionnaire

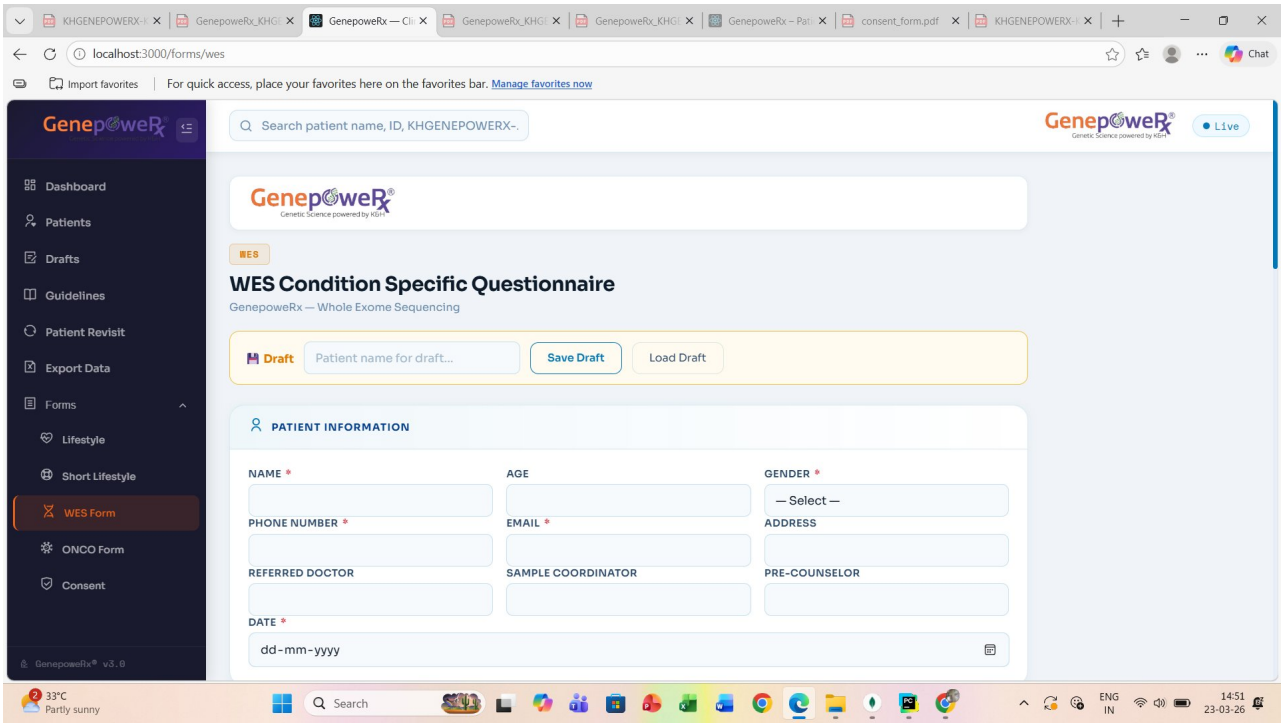

Fig. 9: *The WES (Whole Exome Sequencing) form captures specialized genetic testing information including patient demographics, sample details, referred doctor, sample coordinator, pre-counselor, and date fields.*

3. SYSTEM FEATURES IN DETAIL

3.1 The Five Forms

Form	Purpose	Key Sections
Lifestyle	Full patient intake	Demographics, complaints, medical history, family history, mental health, medications, review of systems, diet, physical exam
Short Lifestyle	Abbreviated intake	Demographics, test selection, medications, family history, mental health, sexual health
WES	Whole Exome Sequencing	Clinical symptoms, age of onset, consanguinity, disease progression, investigations, genetic testing history, pedigree
ONCO	Oncology / Somatic Testing	Cancer type & stage, primary tumor site, pathology report number, family cancer history, diet & lifestyle, FFPE block details
Consent	Informed Consent + Signature	10-point consent declaration, agreement checkbox, canvas digital signature — signature embedded in PDF

3.2 Multi-Visit Tracking

Every form submission is stored as a distinct visit with visitNumber, visitLabel (e.g. "Visit 1"), and isRevisit flag. When a patient submits an already-completed form, the system automatically increments to the next visit number. All previous visits remain intact and accessible for viewing and PDF download. Files uploaded per visit are stored in separate visit folders.

3.3 File Storage Structure

All patient files are organised per patient and per visit under the backend/uploads/ directory:

```
uploads/ KHGENEPOWERX-JOHNDOE/ visit_1/ pdfs/ ← Auto-generated form PDF + consent PDF
uploads/ ← Files uploaded by patient visit_2/ pdfs/ ← Visit 2 PDF uploads/ ← Visit 2
uploaded files
```

4. API REFERENCE

All endpoints served from http://localhost:5000/api/

Analytics & Patients

Method	Endpoint	Description
GET	`/api/analytics`	Dashboard stats — counts, charts, recent submissions
GET	`/api/patients`	All patients (paginated)
GET	`/api/patients/search?q=`	Search by name, ID, phone, email
GET	`/api/patients/:id`	Single patient with forms and drafts
POST	`/api/patients`	Create or find existing patient

Forms & Visits

Method	Endpoint	Description
POST	`/api/forms/submit`	Submit form — multipart/form-data, supports file uploads
POST	`/api/forms/consent`	Submit consent form with digital signature
GET	`/api/forms`	List submissions with optional filters
GET	`/api/history/:patientId`	All visits grouped by form type (for pre-fill)
GET	`/api/visits/:patientId`	Visit summary per form type
GET	`/api/visits/:patientId/:formType/:visitNumber`	Single visit data

PDF Downloads

Method	Endpoint	Description
GET	`/api/pdf/submission/:submissionId`	Download PDF for a specific visit
GET	`/api/pdf/:patientId/:type`	Download latest form or consent PDF

Excel Exports

Method	Endpoint	Description
GET	`/api/export/patients`	All patients registry Excel
GET	`/api/export/form/:formType`	Form-wise Excel (LIFESTYLE/WES/ONCO/etc.)
GET	`/api/export/clinical-insights`	Clinical insights multi-sheet Excel
GET	`/api/export/patient/:patientId`	Single patient all-data Excel

Drafts

Method	Endpoint	Description
POST	`/api/drafts`	Save or update a draft
GET	`/api/drafts`	List drafts with optional search
PUT	`/api/drafts/:id`	Update draft data
DELETE	`/api/drafts/:id`	Delete a draft

5. HOW TO RUN

Terminal 1 — Backend

```
cd C:\Users\perra\PycharmProjects\GenepoweRx_MultiForm_Website\GenepoweRx_MultiForm_Website\backend npm install node server.js
```

Expected:

```
■ Server running on port 5000 ■ MongoDB Connected: localhost (database: genepowerx_internals)
```

Terminal 2 — Frontend

```
cd C:\Users\perra\PycharmProjects\GenepoweRx_MultiForm_Website\GenepoweRx_MultiForm_Website\frontend npm install npm start
```

Browser opens at: http://localhost:3000

Database

Name	genepowerx_internals
Connection	mongodb://localhost:27017/genepowerx_internals
Collections	patients, formsubmissions, drafts

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